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Method and apparatus with memory management based on memory group

Abstract

An electronic device includes: a host processor configured to: generate a memory group allocated to an expanded virtual memory in response to receiving memory usage information; determine a target memory group comprising a page on which a page fault has occurred among memory groups allocated to the expanded virtual memory, based on the memory usage information; and migrate the target memory group from one of a host memory, an accelerator memory, and a storage device, which comprises the target memory group, to either one of the host memory and the accelerator memory in which the page fault has occurred; and an accelerator configured to perform an operation using the accelerator memory, wherein the expanded virtual memory is a virtual single address space using the host memory, the accelerator memory, and the storage device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit under 35 USC § 119(a) of Korean Patent Application No. 10-2023-0010174, filed on Jan. 26, 2023 in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference for all purposes.

BACKGROUND

1. Field

(2) The following description relates to a method and apparatus with memory management based on a memory group.

2. Description of Related Art

(3) A central processing unit (CPU) memory and a graphics processing unit (GPU) memory may be physically separated and divided by a peripheral component interconnect express bus (PCI-Express Bus). Data shared by a CPU and a GPU may need to be allocated to both the CPU memory and the GPU memory. Thus, it may be complicated for a programmer to design or write a program.

(4) When an expanded virtual memory that integrated the CPU memory and the GPU memory is used, it may be easier to design or write the program. The CPU memory and the GPU memory may be physically different memory devices. However, by virtually providing a single address space, an expanded virtual memory space expanded to a CPU memory that is larger than the GPU memory may be used.

SUMMARY

(5) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

(6) In one or more general aspects, an electronic device includes: a host processor configured to: generate a memory group allocated to an expanded virtual memory in response to receiving memory usage information; determine a target memory group comprising a page on which a page fault has occurred among memory groups allocated to the expanded virtual memory, based on the memory usage information; and migrate the target memory group from one of a host memory, an accelerator memory, and a storage device, which comprises the target memory group, to either one of the host memory and the accelerator memory in which the page fault has occurred; and an accelerator configured to perform an operation using the accelerator memory, wherein the expanded virtual memory is a virtual single address space using the host memory, the accelerator memory, and the storage device.

(7) For the generating of the memory group, the host processor may be configured to receive the memory usage information through a hint function configured to input the memory usage information, wherein hint function comprises information about a feature of data included in the generated memory group.

(8) The memory usage information may include a start address of the memory group in the expanded virtual memory, a size of the memory group in the expanded virtual memory, whether the memory group is allocated to the expanded virtual memory, and role information of the memory

group.

(9) The role information of the memory group may be information about a feature of data included in the memory group.

(10) For the migrating of the target memory group, the host processor may be configured to migrate all pages included in the target memory group comprising the page on which the page fault has occurred.

(11) The host processor may be configured to manage the memory groups allocated to the expanded virtual memory with an algorithm configured to perform a range search.

(12) For the managing of the memory groups allocated to the expanded virtual memory with the algorithm configured to perform the range search, the host processor may be configured to determine the target memory group comprising the page on which the page fault has occurred, using the algorithm configured to perform the range search.

(13) For the migrating of the target memory group, the host processor may be configured to, in response to an insufficient space in the host memory to which the target memory group migrates, determine a page that is last accessed among pages included in the host memory and remove all pages included in target memory group to which the page that is last accessed belongs from the host memory.

(14) For the migrating of the target memory group, the host processor may be configured to, in response to an insufficient space in the accelerator memory to which the target memory group migrates, determine a page that is last accessed among pages included in the accelerator memory and remove all pages included in target memory group to which the page that is last accessed belongs from the accelerator memory.

(15) For the migrating of the target memory group, the host processor may be configured to preferentially migrate pages adjacent to the page on which the page fault has occurred among all the pages included in the target memory group.

(16) In one or more general aspects, a processor-implemented method of operating an electronic device includes: generating a memory group allocated to an extended virtual memory in response to receiving memory usage information; determining a target memory group comprising a page on which a page fault has occurred among memory groups allocated to the expanded virtual memory, based on the memory usage information; and migrating the target memory group from one of a host memory, an accelerator memory, and a storage device, which may include the target memory group to either one of the host memory and the accelerator memory in which the page fault has occurred, wherein the expanded virtual memory is a virtual single address space using the host memory, the accelerator memory, and the storage device.

(17) The generating of the memory group may include generating the memory group in response to receiving the memory usage information through a hint function configured to input the memory usage information, wherein hint function may include information about a feature of data included in the generated memory group.

(18) The memory usage information may include a start address of the memory group in the expanded virtual memory, a size of the memory group in the expanded virtual memory, whether the memory group is allocated to the expanded virtual memory, and role information of the memory group.

(19) The role information of the memory group may be information about a feature of data included in the memory group.

(20) The migrating of the target memory group may include migrating all pages included in the target memory group comprising the page on which the page fault has occurred.

(21) The may include managing the memory groups allocated to the expanded virtual memory with an algorithm configured to perform a range search.

(22) The managing of the memory groups allocated to the expanded virtual memory with the algorithm configured to perform the range search may include determining the target memory

group comprising the page on which the page fault has occurred using the algorithm configured to perform the range search.

(23) The migrating of the target memory group may include, in response to an insufficient space in the host memory to which the target memory group migrates, determining a page that is last accessed among pages included in the host memory and removing all pages included in target memory group to which the page that is last accessed belongs from the host memory.

(24) The migrating of the target memory group may include, in response to an insufficient space in the accelerator memory to which the target memory group migrates, determining a page that is last accessed among pages included in the accelerator memory and removing all pages included in target memory group to which the page that is last accessed belongs from the accelerator memory.

(25) The migrating of the target memory group may include preferentially migrating pages adjacent to the page on which the page fault has occurred among all the pages included in the target memory group.

(26) Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates an example of an electronic device.

(2) FIG. 2 illustrates a graph of an example of increase of memory usage of deep learning models.

(3) FIG. 3 illustrates a graph of an example of comparison of latency time in an expanded virtual memory and a general memory.

(4) FIG. 4 illustrates an example of memory allocation of an application.

(5) FIGS. 5A and 5B illustrate an example of allocation of memory groups.

(6) FIG. 6 illustrates an example of managing a structure.

(7) FIG. 7 illustrates an example of migration of memory groups.

(8) FIG. 8 illustrates an example of removing a memory group.

(9) FIG. 9 illustrates an example of an operation of an electronic device.

(10) Throughout the drawings and the detailed description, unless otherwise described or provided, the same drawing reference numerals will be understood to refer to the same elements, features, and structures. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

(11) The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. However, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be apparent after an understanding of the disclosure of this application. For example, the sequences of operations described herein are merely examples, and are not limited to those set forth herein, but may be changed as will be apparent after an understanding of the disclosure of this application, with the exception of operations necessarily occurring in a certain order. Also, descriptions of features that are known after an understanding of the disclosure of this application may be omitted for increased clarity and conciseness.

(12) Although terms, such as “first,” “second,” and “third”, or A, B, (a), (b), and the like may be used herein to describe various members, components, regions, layers, or sections, these members, components, regions, layers, or sections are not to be limited by these terms. Each of these terminologies is not used to define an essence, order, or sequence of corresponding members, components, regions, layers, or sections, for example, but used merely to distinguish the corresponding members, components, regions, layers, or sections from other members,

components, regions, layers, or sections. Thus, a first member, component, region, layer, or section referred to in the examples described herein may also be referred to as a second member, component, region, layer, or section without departing from the teachings of the examples.

(13) Throughout the specification, when a component or element is described as being “connected to,” “coupled to,” or “joined to” another component or element, it may be directly (e.g., in contact with the other component or element) “connected to,” “coupled to,” or “joined to” the other component or element, or there may reasonably be one or more other components or elements intervening therebetween. When a component or element is described as being “directly connected to,” “directly coupled to,” or “directly joined to” another component or element, there can be no other elements intervening therebetween. Likewise, expressions, for example, “between” and “immediately between” and “adjacent to” and “immediately adjacent to” may also be construed as described in the foregoing.

(14) The terminology used herein is for describing various examples only and is not to be used to limit the disclosure. The articles “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. As used herein, the term “and/or” includes any one and any combination of any two or more of the associated listed items. As non-limiting examples, terms “comprise” or “comprises,” “include” or “includes,” and “have” or “has” specify the presence of stated features, numbers, operations, members, elements, and/or combinations thereof, but do not preclude the presence or addition of one or more other features, numbers, operations, members, elements, and/or combinations thereof.

(15) Unless otherwise defined, all terms used herein including technical or scientific terms have the same meanings as those generally understood by one of ordinary skill in the art to which this disclosure pertains and based on an understanding of the disclosure of the present disclosure. Terms, such as those defined in commonly used dictionaries, should be construed to have meanings matching with contextual meanings in the relevant art and the present disclosure, and are not to be construed as an ideal or excessively formal meaning unless otherwise defined herein.

(16) The phrases “at least one of A, B, and C”, “at least one of A, B, or C”, and the like are intended to have disjunctive meanings, and these phrases “at least one of A, B, and C”, “at least one of A, B, or C”, and the like also include examples where there may be one or more of each of A, B, and/or C (e.g., any combination of one or more of each of A, B, and C), unless the corresponding description and embodiment necessitates such listings (e.g., “at least one of A, B, and C”) to be interpreted to have a conjunctive meaning. The use of the term “may” herein with respect to an example or embodiment, e.g., as to what an example or embodiment may include or implement, means that at least one example or embodiment exists where such a feature is included or implemented, while all examples are not limited thereto.

(17) Hereinafter, examples will be described in detail with reference to the accompanying drawings. When describing the examples with reference to the accompanying drawings, like reference numerals refer to like components and a repeated description related thereto will be omitted.

(18) FIG. 1 illustrates an example of an electronic device.

(19) Referring to FIG. 1, an electronic device **100** including a host processor **110** (e.g., one or more processors), an accelerator **120** (e.g., one or more accelerators), a host memory **140** (e.g., one or more memories), a storage device **150** (e.g., one or more storage devices), and an accelerator memory **160** (e.g., one or more accelerator memories) is illustrated. The host processor **110**, the accelerator **120**, the host memory **140**, the storage device **150**, and the accelerator memory **160** may communicate with each other via a bus, a network-on-chip (NoC), a peripheral component interconnect express (PCIe), and the like. Only components related to the present example are illustrated in the electronic device **100** of FIG. 1. Thus, it will be apparent to one of ordinary skill in the art, with an understanding of the disclosure of the present disclosure, that the electronic device **100** may further include general-purpose components other than the components illustrated in FIG.

1.

(20) The host processor **110** may perform overall functions to control the electronic device **100**. The host processor **110** may generally control the electronic device **100** by executing programs and/or instructions stored in memories. The host processor **110** may be implemented as a central processing unit (CPU), a graphics processing unit (GPU), an application processor (AP), and the like but is not limited thereto.

(21) The electronic device **100** may include the accelerator **120** (or a hardware accelerator) for operations. The accelerator **120** may process tasks that are, due to a characteristic of the tasks, more efficiently processed by a separate dedicated processor (e.g., the accelerator **120**) than by the host processor **110** configured for general purpose. One or more processing elements (PEs) included in the accelerator **120** may be used. For example, the accelerator **120** may correspond to a neural processing unit (NPU), a tensor processing unit (TPU), a digital signal processor (DSP), a GPU, a neural engine, and the like.

(22) The host memory **140**, the storage device **150**, and the accelerator memory **160** may be hardware configured to store data that is processed or may be processed by the electronic device **100**. In addition, the host memory **140**, the storage device **150**, and the accelerator memory **160** may store an application, a driver, and the like to be executed by the electronic device **100**. The host memory **140**, the storage device **150**, and the accelerator memory **160** may include a volatile memory, such as dynamic random-access memory (DRAM), and/or a non-volatile memory.

(23) The host memory **140** may be a memory of the host processor **110**. Data may be allocated to the host memory **140** for the host processor **110** to perform an operation. For example, when the host processor **110** is a CPU, the host memory **140** may be DRAM as the CPU memory. For example, the host memory **140** may be a memory connected to the electronic device **100** through a computer express link (CXL) interface. In a non-limiting example, the host memory **140** may be or include a non-transitory computer-readable storage medium storing instructions that, when executed by the host processor **110**, configure the host processor **110** to perform any one, any combination, or all of operations and methods of the host processor **110** disclosed herein with reference to FIGS. 1-9.

(24) The accelerator memory **160** may be a memory of the accelerator **120**. Data may be allocated to the accelerator memory **160** for the accelerator **120** to perform an operation. For example, when the accelerator **120** is the GPU, the accelerator memory **160** may be a GPU memory.

(25) The storage device **150** may be hardware configured to store data processed by the electronic device **100** and may store an application, a driver, and the like to be executed by the electronic device **100**. Space of the storage device **150** may be larger than the space of the host memory **140** and the accelerator memory **160**. The storage device **150** may include a solid-state drive (SSD), a hard disk drive (HDD), and the like but is not limited thereto.

(26) An expanded virtual memory **130** may be a virtual single address space using the host memory **140**, the storage device **150**, and the accelerator memory **160**. For example, the expanded virtual memory **130** may be an expanded virtual memory integrating the host memory **140**, the storage device **150**, and the accelerator memory **160**. The size of the expanded virtual memory **130** may be the sum of the size of the host memory **140**, the size of the accelerator memory **160**, and the size of the free space of the storage device **150**. The expanded virtual memory **130** may use the free space of the storage device **150**, like the host memory **140** and/or the accelerator memory **160**.

(27) Thus, as the size of the expanded virtual memory **130** may be expanded to the free space of the storage device **150**, the electronic device **100** of one or more embodiments may more easily execute an application when executing the application, for example, uses a memory capacity larger than the size of the host memory **140** and the accelerator memory **160**.

(28) FIG. 2 illustrates a graph of an example of increase of memory usage of deep learning models.

(29) Referring to FIG. 2, a graph **200** showing an increase of the memory usage of the deep learning models due to technological development in the last 10 years is illustrated. Referring to the

graph **200**, it is shown that the deep learning models have developed in a way that the size of the models increases in order to improve accuracy of learning. Accordingly, a memory capacity used for learning of a deep learning model may gradually increase. For example, GPT-3, one of the language models, may include around 175 billion learning parameters, and the memory capacity used for learning through GPT-3 may exceed 1000 GB. Thus, the expanded virtual memory **130** of FIG. **1** may support a significantly large virtual memory by supporting up to the size of the free space of the storage device **150**. Therefore, by using the expanded virtual memory **130**, the electronic device **100** of one or more embodiments may learn with and execute a significantly large deep learning model.

(30) FIG. **3** illustrates a graph **300** of an example of comparison of latency time in an expanded virtual memory and a general memory.

(31) As described with reference to FIG. **1**, the expanded virtual memory **130** using the host memory **140**, the storage device **150**, and the accelerator memory **160** may train or execute the deep learning model of a large size by providing a significantly large expanded virtual memory **130** expanded to the storage device **150**. However, software overhead incurred to support the expanded virtual memory **130** may be significantly large.

(32) The graph **300** may be a graph showing a comparison of the latency time of the expanded virtual memory **130** (expanded to the host memory **140** and the accelerator memory **160**, for example) and the latency time of the general memory. Referring to the graph **300**, it is shown that the latency time of the expanded virtual memory **130** is higher than the latency time of the general memory.

(33) The latency time of the expanded virtual memory **130** may be higher than the latency time of the general memory when the expanded virtual memory **130** operates based on demand paging. When the expanded virtual memory **130** of a typical electronic device accesses the host memory **140**, the accelerator memory **160**, or the storage device **150**, when a page fault occurs, a page on which the page fault has occurred may be placed to the memory the expanded virtual memory **130** accessed by a page fault handler. Due to the above operation, the latency time of the expanded virtual memory **130** of the typical electronic device may be high. In contrast, to reduce the latency time of the expanded virtual memory **130**, the electronic device **100** of one or more embodiments may implement a method of prefetching a page. For such prefetching, the electronic device **100** may determine how a virtual memory (e.g., the expanded virtual memory **130**) is used. Hereinafter, a method for a deep learning application **410** to use a virtual memory is described.

(34) FIG. **4** illustrates an example of memory allocation of an application.

(35) The deep learning application **410** may, to reduce overhead, manage the memory by making a dedicated memory pool rather than requesting allocation and/or release of the memory from the expanded virtual memory **130**.

(36) The deep learning application **410**, when executed, may manage the memory through a framework **420**. The deep learning application **410**, when executed, may be allocated to the memory to use through the framework **420** and manage the memory in the memory pool. The deep learning application **410** may request allocation when the memory is determined to be used and request release when the memory usage is finished. However, the framework **420** may process the request internally and not request the allocation and release of the memory from the extended virtual memory **130**. The framework **420** may often request the allocation and release of the memory from the expanded virtual memory **130** to maintain the internal memory pool.

(37) Thus, the expanded virtual memory **130** of the typical electronic device may not determine how the deep learning application **410** uses the memory. In contrast, the expanded virtual memory **130** of the electronic device **100** of one or more embodiments may determine how the deep learning application **410** uses the memory, thereby reducing the latency time by prefetching a page. Hereinafter, an example method of notifying the expanded virtual memory **130** of how the deep learning application **410** uses the memory is described.

(38) FIGS. 5A and 5B illustrate an example of allocation of memory groups.

(39) Referring to FIG. 5A, a hint function **500** is illustrated. The hint function **500** may be a function that is configured to receive memory usage information of a memory group input from a user. The memory group may be a unit for managing the memory allocated to the expanded virtual memory **130** by the hint function **500**. The memory group may include a plurality of pages. For example, the plurality of pages may be managed in the memory group.

(40) The hint function **500** may be an application programming interface (API). The hint function **500** may be an expansion of an existing function to receive the memory usage information input from the user. Alternatively, the hint function **500** may be a newly defined function to receive the memory usage information input from the user.

(41) The hint function **500** of FIG. 5A may be a function that expands cudaMemAdvise, which is an API. For example, the hint function **500** may receive an address **510** (e.g., addr of FIG. 5A), a length **520** (e.g., len of FIG. 5A), an advice **530** (e.g., advice of FIG. 5A), and a type **540** (e.g., type of FIG. 5A) input from the user.

(42) For example, the address **510** may represent a start address of the memory group in the expanded virtual memory **130**. The length **520** may represent the size of the memory group in the expanded virtual memory **130**.

(43) The advice **530** may represent whether the memory group is allocated to the expanded virtual memory **130**. When the advice **530** is set, the memory group may be allocated to the expanded virtual memory **130**, and when the advice **530** is unset, the memory group may be removed from the expanded virtual memory **130**.

(44) The type **540** may represent role information of the memory group. The role information may be information about a feature of data included in the memory group. For example, when the data included in the memory group is a matrix, the role information may include information about sparsity or density of the matrix. For example, the role information may include information about when the memory group is set and when the memory group is unset.

(45) The memory usage information may include the address **510**, the length **520**, the advice **530**, and the type **540** described above. Thus, the memory usage information may be a kind of hint on how the memory group is used. As a result, the expanded virtual memory **130** may obtain the hint on the data included in the memory group through the memory usage information received as an input from the user.

(46) Referring to FIG. 5B, memory groups allocated to the expanded virtual memory **130** are illustrated. The allocated memory groups may be managed as a structure. For example, referring to FIG. 5B, the memory groups allocated to the expanded virtual memory **130** may be managed as a vm_group structure. However, the vm_group structure is only an example and the memory groups are not limited thereto.

(47) Here, “Start” may correspond to the address **510**. “end” may correspond to the sum of the address **510** and the length **520**. “type” may correspond to the type **540**.

(48) Different memory groups may include different pages. In addition, different memory groups may include different numbers of pages.

(49) FIG. 6 illustrates an example of managing a structure.

(50) The structure to manage the memory groups may be quickly searched for using a specific virtual address. When the page fault occurs, the structure to manage the memory groups may be quickly searched for to perform page fault handling using information included in the structure.

(51) When the virtual address of the page on which the page fault has occurred is provided, as the structure to manage the memory group including the virtual address is to be found, an algorithm configured to perform a range search may be used to manage the structure.

(52) Referring to FIG. 6, a red-black tree **600**, which is an example of the algorithm configured to perform the range search, is illustrated. The red-black tree **600** is only an example of the algorithm that may be used to manage the structure, and the algorithm is not limited thereto. Thus, any

algorithm configured to perform the range search may be used to find the structure including the page on which the page fault has occurred.

(53) FIG. 7 illustrates an example of migration of memory groups.

(54) Referring to FIG. 7, a method of migrating memory groups when a page fault occurs is described.

(55) The expanded virtual memory **130**, a kind of virtual memory, may be a physically different memory device but configured to virtually provide a single address space, thus providing convenience in programming. The expanded virtual memory **130** may operate based on demand paging.

(56) In an example, the host processor **110** may access a page included in a memory group 1 for an operation. The host processor **110** may perform the operation using data in the host memory **140** only. Referring to the host memory **140**, the host memory **140** may include a memory group 2 but may not include the memory group 1. Thus, the page fault may occur when the memory group 1 is included in the expanded virtual memory **130** but the page included in the memory group 1 is not included in the host memory **140**.

(57) The storage device **150** and/or the accelerator memory **160** may include the memory group 1. Thus, when the page fault occurs, the memory group 1 of the storage device **150** or the accelerator memory **160** may migrate to the host memory **140**. When the memory group 1 migrates to the host memory **140**, the host processor **110** may access the memory group 1 of the host memory **140**.

(58) Here, not only the page on which the page fault has occurred may migrate, but a target memory group including the page may migrate. An example method of finding the target memory group including the page on which the page fault has occurred is described above with reference to FIG. 6. For example, the structure including the page may be found using the address of the page on which the page fault has occurred. The structure may be the target memory group including the page on which the page fault has occurred.

(59) When the target memory group including the page migrates, all pages included in the target memory group including the page on which the page fault has occurred may migrate. The page fault may be predicted to occur on the pages included in the target memory group including the page on which the page fault has occurred after the page fault has occurred. Thus, by migrating all the pages included in the target memory group, the electronic device of one or more embodiments may obtain an effect of prefetching pages on which the page fault has not yet occurred but is predicted to occur.

(60) Accordingly, the electronic device of one or more embodiments may reduce the number of the occurrence of page faults by migrating all the pages included in the target memory group, but handling time may increase when the size of target memory group is large when the pages migrate in memory group units. When the handling time increases, the deep learning application **410** may wait in standby status. Thus, by preferentially migrating pages adjacent to the page on which the page fault has occurred, the electronic device of one or more embodiments may reduce the handling time. For example, by synchronously migrating the pages adjacent to the page on which the page fault has occurred and asynchronously processing pages not adjacent to the page on which the page fault has occurred in the background, the electronic device of one or more embodiments may reduce the handling time.

(61) The description above may be applied not only when the page fault occurs in the host memory **140** and the target memory group migrates in the storage device **150** or the accelerator memory **160** but also when the page fault occurs in the accelerator memory **160** and the target memory group migrates in the storage device **150** or the host memory **140**.

(62) An example policy to remove the memory group when a memory space is insufficient even though the migration is performed because the page fault has occurred is described below.

(63) FIG. 8 illustrates an example of removing a memory group.

(64) Referring to FIG. 8, the host memory **140** is illustrated. However, it will be apparent to one of

ordinary skill in the art that the removing of the memory group may be applied to not only the host memory **140** but also to other memories of the electronic device **100** (e.g., the accelerator memory **160**).

(65) The host memory **140** may include a plurality of memory groups. The plurality of memory groups may be managed by two least recently used (LRU) lists. The two LRU lists may be an active list **810** and an inactive list **820**. The active list **810** may manage hot memory groups with a lot of accesses. The inactive list **820** may manage cold memory groups with few accesses. The active list **810** and the inactive list **820** may each include a plurality of memory groups.

(66) When a new memory group **830** is input to the host memory **140** for the first time, the new memory group **830** may be added to the inactive list **820**. Among the memory groups included in the inactive list **820**, memory groups repeatedly accessed may be moved to the active list **810**. On the other hand, when there are too many memory groups in the active list **810** (e.g., when there are more memory groups in the active list **810** than the active list **810** is configured to manage, and/or when there the number of memory groups in the active list **810** is the maximum number of memory groups the active list **810** is configured to manage and another memory group is determined to be moved to the active list **810**), a last accessed memory group **811** may be moved to the inactive list **820**. In an example, when there are too many memory groups in the active list **810**, the host processor **110** may move at least one of the memory groups in the active list **810** to the inactive list **820**.

(67) When there are too many memory groups in the inactive list **820** (e.g., when there are more memory groups in the inactive list **820** than the inactive list **820** is configured to manage, and/or when there the number of memory groups in the inactive list **820** is the maximum number of memory groups the inactive list **820** is configured to manage and another memory group is determined to be moved to the active list **810**), a last accessed memory group **821** of the inactive list **820** may be removed. In an example, when there are too many memory groups in the inactive list **820**, the host processor **110** may move the last accessed memory group **821** of the inactive list **820** to another memory device (e.g., the storage device **150**, the expanded virtual memory **130**, the accelerator memory **160**, and the like) and remove the last accessed memory group **821** of the inactive list **820** from the host memory **140**.

(68) The host processor **110** may move part of the last accessed memory group **821** of the inactive list **820** to another memory device and remove the part of the last accessed memory group **821** of the inactive list **820** from the host memory **140**. Then, the host processor **110** may move the rest of the last accessed memory group **821** of the inactive list **820** to another memory device and remove the rest of the last accessed memory group **821** of the inactive list **820** from the host memory **140**.

(69) FIG. **9** illustrates an example of an operation of an electronic device.

(70) Each operation may be performed sequentially, but not necessarily sequentially performed. For example, an order of each operation may be changed and at least two operations may be performed in parallel. Operations **910** to **930** may be performed by the electronic device **100** of FIG. **1**. For example, the host processor **110** of the electronic device **100** may perform the below operations.

(71) In operation **910**, a host processor may generate a memory group allocated to an expanded virtual memory in response to receiving memory usage information input from a user.

(72) In operation **920**, the host processor may determine a target memory group including a page on which a page fault has occurred among memory groups allocated to the expanded virtual memory, based on the memory usage information.

(73) In operation **930**, the host processor may migrate the target memory group from one of a host memory, an accelerator memory, and a storage device, which includes the target memory group to either the host memory or the accelerator memory in which the page fault has occurred.

(74) The electronic devices, host processors, accelerators, host memories, storage devices, accelerator memories, electronic device **100**, host processor **110**, accelerator **120**, host memory **140**, storage device **150**, accelerator memory **160**, and other apparatuses, devices, units, modules,

and components disclosed and described herein with respect to FIGS. 1-9 are implemented by or representative of hardware components. As described above, or in addition to the descriptions above, examples of hardware components that may be used to perform the operations described in this application where appropriate include controllers, sensors, generators, drivers, memories, comparators, arithmetic logic units, adders, subtractors, multipliers, dividers, integrators, and any other electronic components configured to perform the operations described in this application. In other examples, one or more of the hardware components that perform the operations described in this application are implemented by computing hardware, for example, by one or more processors or computers. A processor or computer may be implemented by one or more processing elements, such as an array of logic gates, a controller and an arithmetic logic unit, a digital signal processor, a microcomputer, a programmable logic controller, a field-programmable gate array, a programmable logic array, a microprocessor, or any other device or combination of devices that is configured to respond to and execute instructions in a defined manner to achieve a desired result. In one example, a processor or computer includes, or is connected to, one or more memories storing instructions or software that are executed by the processor or computer. Hardware components implemented by a processor or computer may execute instructions or software, such as an operating system (OS) and one or more software applications that run on the OS, to perform the operations described in this application. The hardware components may also access, manipulate, process, create, and store data in response to execution of the instructions or software. For simplicity, the singular term “processor” or “computer” may be used in the description of the examples described in this application, but in other examples multiple processors or computers may be used, or a processor or computer may include multiple processing elements, or multiple types of processing elements, or both. For example, a single hardware component or two or more hardware components may be implemented by a single processor, or two or more processors, or a processor and a controller. One or more hardware components may be implemented by one or more processors, or a processor and a controller, and one or more other hardware components may be implemented by one or more other processors, or another processor and another controller. One or more processors, or a processor and a controller, may implement a single hardware component, or two or more hardware components. As described above, or in addition to the descriptions above, example hardware components may have any one or more of different processing configurations, examples of which include a single processor, independent processors, parallel processors, single-instruction single-data (SISD) multiprocessing, single-instruction multiple-data (SIMD) multiprocessing, multiple-instruction single-data (MISD) multiprocessing, and multiple-instruction multiple-data (MIMD) multiprocessing.

(75) The methods illustrated in FIGS. 1-9 that perform the operations described in this application are performed by computing hardware, for example, by one or more processors or computers, implemented as described above implementing instructions or software to perform the operations described in this application that are performed by the methods. For example, a single operation or two or more operations may be performed by a single processor, or two or more processors, or a processor and a controller. One or more operations may be performed by one or more processors, or a processor and a controller, and one or more other operations may be performed by one or more other processors, or another processor and another controller. One or more processors, or a processor and a controller, may perform a single operation, or two or more operations.

(76) Instructions or software to control computing hardware, for example, one or more processors or computers, to implement the hardware components and perform the methods as described above may be written as computer programs, code segments, instructions or any combination thereof, for individually or collectively instructing or configuring the one or more processors or computers to operate as a machine or special-purpose computer to perform the operations that are performed by the hardware components and the methods as described above. In one example, the instructions or software include machine code that is directly executed by the one or more processors or

computers, such as machine code produced by a compiler. In another example, the instructions or software includes higher-level code that is executed by the one or more processors or computer using an interpreter. The instructions or software may be written using any programming language based on the block diagrams and the flow charts illustrated in the drawings and the corresponding descriptions herein, which disclose algorithms for performing the operations that are performed by the hardware components and the methods as described above.

(77) The instructions or software to control computing hardware, for example, one or more processors or computers, to implement the hardware components and perform the methods as described above, and any associated data, data files, and data structures, may be recorded, stored, or fixed in or on one or more non-transitory computer-readable storage media, and thus, not a signal per se. As described above, or in addition to the descriptions above, examples of a non-transitory computer-readable storage medium include one or more of any of read-only memory (ROM), random-access programmable read only memory (PROM), electrically erasable programmable read-only memory (EEPROM), random-access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), flash memory, non-volatile memory, CD-ROMs, CD-Rs, CD+Rs, CD-RWs, CD+RWs, DVD-ROMs, DVD-Rs, DVD+Rs, DVD-RWs, DVD+RWs, DVD-RAMs, BD-ROMs, BD-Rs, BD-R LTHs, BD-REs, blue-ray or optical disk storage, hard disk drive (HDD), solid state drive (SSD), flash memory, a card type memory such as multimedia card micro or a card (for example, secure digital (SD) or extreme digital (XD)), magnetic tapes, floppy disks, magneto-optical data storage devices, optical data storage devices, hard disks, solid-state disks, and any other device that is configured to store the instructions or software and any associated data, data files, and data structures in a non-transitory manner and provide the instructions or software and any associated data, data files, and data structures to one or more processors or computers so that the one or more processors or computers can execute the instructions. In one example, the instructions or software and any associated data, data files, and data structures are distributed over network-coupled computer systems so that the instructions and software and any associated data, data files, and data structures are stored, accessed, and executed in a distributed fashion by the one or more processors or computers.

(78) While this disclosure includes specific examples, it will be apparent after an understanding of the disclosure of this application that various changes in form and details may be made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner, and/or replaced or supplemented by other components or their equivalents.

(79) Therefore, in addition to the above and all drawing disclosures, the scope of the disclosure is also inclusive of the claims and their equivalents, i.e., all variations within the scope of the claims and their equivalents are to be construed as being included in the disclosure.

Claims

1. An electronic device comprising: a host processor configured to: generate, in response to receiving memory usage information, a memory group allocated to an expanded virtual memory, the expanded virtual memory is a virtual single address space using a host memory, an accelerator memory, and a storage device; determine, based on the memory usage information, from among memory groups allocated to the expanded virtual memory, a target memory group comprising a page on which a page fault has occurred; and migrate the target memory group from one of the host memory, the accelerator memory, and the storage device that includes the target memory group, to

- either one of the host memory and the accelerator memory in which the page fault has occurred; and an accelerator configured to perform an operation using the accelerator memory, wherein the memory groups comprise a first memory group and a second memory group having different allocations of at least one of the host memory, the accelerator memory, and the storage device.
2. The electronic device of claim 1, wherein, for the generating of the memory group, the host processor is configured to receive the memory usage information through a hint function configured to input the memory usage information, wherein hint function comprises information about a feature of data included in the generated memory group.
 3. The electronic device of claim 1, wherein the memory usage information comprises a start address of the memory group in the expanded virtual memory, a size of the memory group in the expanded virtual memory, whether the memory group is allocated to the expanded virtual memory, and role information of the memory group.
 4. The electronic device of claim 3, wherein the role information of the memory group is information about a feature of data included in the memory group.
 5. The electronic device of claim 1, wherein, for the migrating of the target memory group, the host processor is configured to migrate all pages included in the target memory group comprising the page on which the page fault has occurred.
 6. The electronic device of claim 1, wherein the host processor is configured to manage the memory groups allocated to the expanded virtual memory with an algorithm configured to perform a range search.
 7. The electronic device of claim 6, wherein, for the managing of the memory groups allocated to the expanded virtual memory with the algorithm configured to perform the range search, the host processor is configured to determine the target memory group comprising the page on which the page fault has occurred, using the algorithm configured to perform the range search.
 8. The electronic device of claim 1, wherein, for the migrating of the target memory group, the host processor is configured to, in response to an insufficient space in the host memory to which the target memory group migrates, determine a page that is last accessed among pages included in the host memory and remove all pages included in target memory group to which the page that is last accessed belongs from the host memory.
 9. The electronic device of claim 1, wherein, for the migrating of the target memory group, the host processor is configured to, in response to an insufficient space in the accelerator memory to which the target memory group migrates, determine a page that is last accessed among pages included in the accelerator memory and remove all pages included in target memory group to which the page that is last accessed belongs from the accelerator memory.
 10. The electronic device of claim 1, wherein, for the migrating of the target memory group, the host processor is configured to preferentially migrate pages adjacent to the page on which the page fault has occurred among all the pages included in the target memory group.
 11. A processor-implemented method of operating an electronic device, the method comprising: generating, in response to receiving memory usage information, a memory group allocated to an expanded virtual memory, the expanded virtual memory is a virtual single address space using a host memory, an accelerator memory, and a storage device; determining, based on the memory usage information, from among memory groups allocated to the expanded virtual memory, a target memory group comprising a page on which a page fault has occurred; and migrating the target memory group from one of the host memory, the accelerator memory, and the storage device that includes the target memory group, to either one of the host memory and the accelerator memory in which the page fault has occurred, wherein the memory groups comprise a first memory group and a second memory group having different allocations of at least one of the host memory, the accelerator memory, and the storage device.
 12. The method of claim 11, wherein the generating of the memory group comprises generating the memory group in response to receiving the memory usage information through a hint function

configured to input the memory usage information, wherein hint function comprises information about a feature of data included in the generated memory group.

13. The method of claim 11, wherein the memory usage information comprises a start address of the memory group in the expanded virtual memory, a size of the memory group in the expanded virtual memory, whether the memory group is allocated to the expanded virtual memory, and role information of the memory group.

14. The method of claim 13, wherein the role information of the memory group is information about a feature of data included in the memory group.

15. The method of claim 11, wherein the migrating of the target memory group comprises migrating all pages included in the target memory group comprising the page on which the page fault has occurred.

16. The method of claim 11, further comprising managing the memory groups allocated to the expanded virtual memory with an algorithm configured to perform a range search.

17. The method of claim 16, wherein the managing of the memory groups allocated to the expanded virtual memory with the algorithm configured to perform the range search comprises determining the target memory group comprising the page on which the page fault has occurred using the algorithm configured to perform the range search.

18. The method of claim 11, wherein the migrating of the target memory group comprises, in response to an insufficient space in the host memory to which the target memory group migrates, determining a page that is last accessed among pages included in the host memory and removing all pages included in target memory group to which the page that is last accessed belongs from the host memory.

19. The method of claim 11, wherein the migrating of the target memory group comprises, in response to an insufficient space in the accelerator memory to which the target memory group migrates, determining a page that is last accessed among pages included in the accelerator memory and removing all pages included in target memory group to which the page that is last accessed belongs from the accelerator memory.

20. The method of claim 11, wherein the migrating of the target memory group comprises preferentially migrating pages adjacent to the page on which the page fault has occurred among all the pages included in the target memory group.
